



Supermarket Billing System

Introduction

- *This project is a simple supermarket billing system created using Python.*
- *It helps calculate total cost, discount, and prints an itemized bill.*





Purpose of the Project

- *Automate billing operations*
- *Reduce manual errors*
- *Provide beginners with practical Python experience*
- *Demonstrate real-life programming logic*

Features

- *Multiple Item Processing:* Allows users to input multiple items with quantities in a single transaction.
- *Discount Application:* Automatically applies discounts based on the total amount:
 - *Above ₹5000 → 20%*
 - *Above ₹2000 → 10%*
 - *Above ₹1000 → 5%*
 - *Below ₹1000 → No discount*
- *Itemized Receipt:* Provides a detailed receipt showing each item, price, quantity, subtotal, discount, and final total.
- *Continuous Input:* Processes multiple transactions for consecutive customers in a single run.

How It Works

Item Selection:

- *A list of available products and prices is displayed.*
- *Customers enter the item name and quantity. The system calculates the subtotal and adds the item to the cart.*

Discount Calculation:

- *The system applies discounts based on the total bill amount:*
- *20% discount for totals up to ₹5000*
- *10% discount for totals up to ₹2000*
- *5% discount for totals up to ₹1000*
- *0% discount for Below ₹1000*

Receipt Generation:

An itemized receipt is generated, displaying:

- *Store information, date, and time*
- *Itemized details (name, price, quantity)*
- *Total bill, discount applied, and the final bill after the discount*

Multiple Transactions:

- *The program can handle consecutive customers in a single run, asking after each receipt whether there is another customer in the queue.*

Code

```
[2]: #Supermarket Billing System
print("===== Supermarket Management System =====")
item_name = []
item_quantity = []
item_price = []
item_total=[]

while True:
    name = input("Enter the item name: ")
    quantity = int(input("Enter the item Quantity: "))
    price = float(input("Enter the item price:"))
    total = quantity * price

    item_name.append(name)
    item_quantity.append(quantity)
    item_price.append(price)
    item_total.append(total)

    choice = input("Do you want to add more items? (yes/no): ").lower( )
    if choice == 'no':
        break
```

```
print("\n===== Bill Invoice =====")
print("{:<15} {:<10} {:<10} {:<10}".format("Name", "Quantity", "Price", "Total"))
print("-----")
subtotal = sum(item_total)

if subtotal >= 5000:
    discount = subtotal * 0.20
elif subtotal >= 2500:
    discount = subtotal * 0.10
elif subtotal >= 1000:
    discount = subtotal * 0.05
else:
    discount = 0

final_total = subtotal - discount

for i in range(len(item_name)):
    print("{:<15} {:<10} {:<10.2f} {:<10.2f}".format(item_name[i], item_quantity[i], item_price[i], item_total[i]))

print("-----")
print(f"Subtotal: ₹{subtotal:.2f}")
print(f"Discount: ₹{discount:.2f}")
print(f"Final Total: ₹{final_total:.2f}")
print("Thank you for shopping with us!")
```

Result

===== Supermarket Management System =====

Enter the item name: Daal
Enter the item Quantity: 5
Enter the item price: 300
Do you want to add more items? (yes/no): yes
Enter the item name: Aata
Enter the item Quantity: 3
Enter the item price: 79
Do you want to add more items? (yes/no): yes
Enter the item name: Oil
Enter the item Quantity: 15
Enter the item price: 458
Do you want to add more items? (yes/no): yes
Enter the item name: Red chilli powder
Enter the item Quantity: 353
Enter the item price: 123
Do you want to add more items? (yes/no): yes
Enter the item name: Rice
Enter the item Quantity: 8
Enter the item price: 90
Do you want to add more items? (yes/no): yes
Enter the item name: Salt
Enter the item Quantity: 4
Enter the item price: 50
Do you want to add more items? (yes/no): no

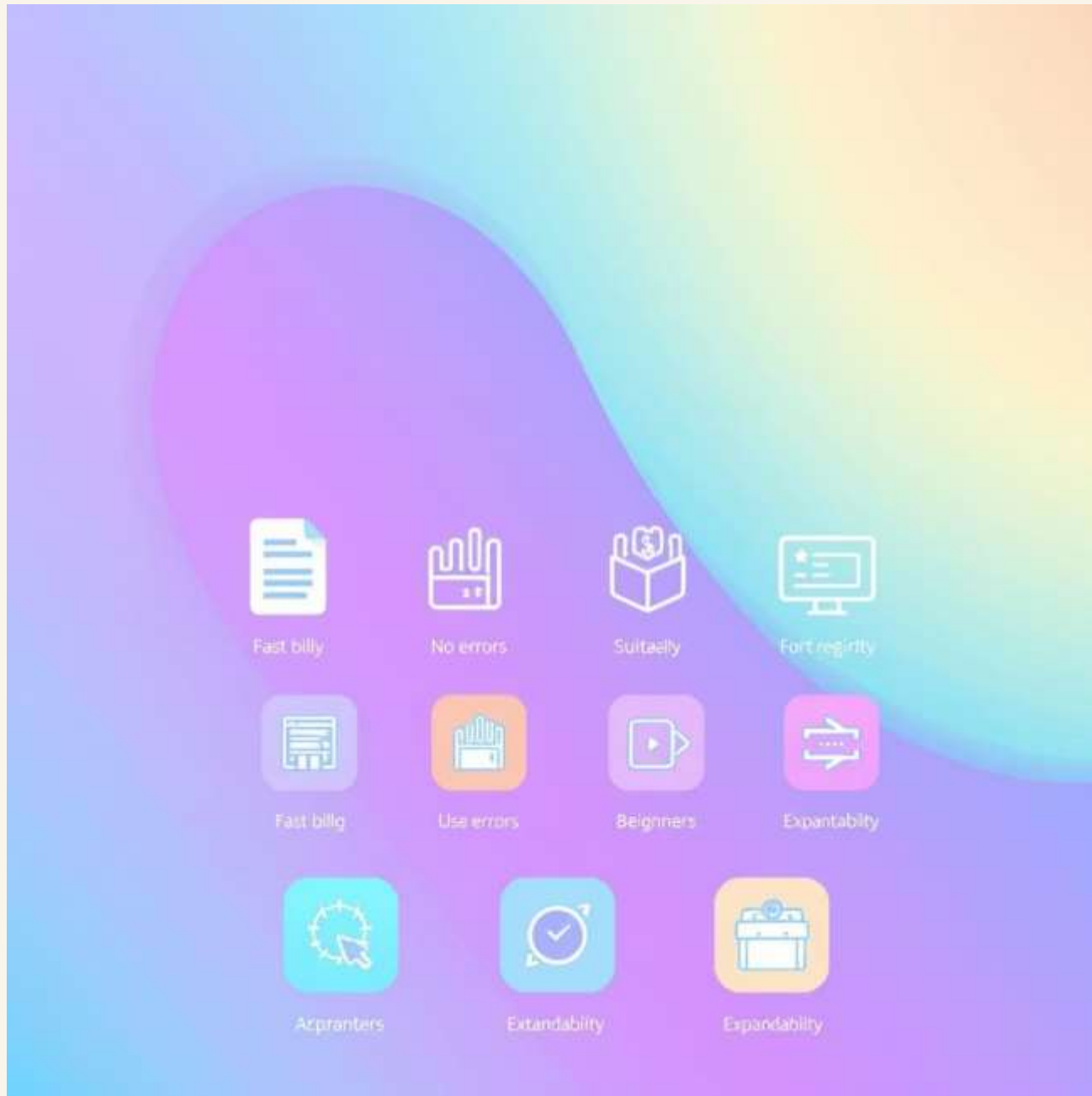
===== Bill Invoice =====

Name	Quantity	Price	Total
Daal	5	300.00	1500.00
Aata	3	79.00	237.00
Oil	15	458.00	6870.00
Red chilli powder	353	123.00	43419.00
Rice	8	90.00	720.00
Salt	4	50.00	200.00

Subtotal: ₹52946.00
Discount: ₹10589.20
Final Total: ₹42356.80
Thank you for shopping with us!

Advantages

- *Fast billing*
- *No errors*
- *Good for beginners*
- *Expandable project*



Conclusion

The Supermarket Billing System successfully streamlines the entire checkout and billing process by automating tasks that were previously manual and time-consuming. Through the use of Python, the system efficiently manages product details, calculates totals, applies discounts, and generates accurate bills. This not only reduces human errors but also improves customer experience by speeding up billing operations.

The project demonstrates how technology can enhance everyday business activities, especially in retail environments where accuracy and efficiency are crucial. Overall, the system is scalable, user-friendly, and can be further improved by adding features such as inventory management, barcode scanning, and database integration. The successful completion of this project highlights the practical application of programming skills in solving real-world problems.

Thank You

By Sanyam Sharma

